

SRINIVAS UNIVERSITY
INSTITUTE OF ENGINEERING & TECHNOLOGY
MUKKA, MANGALURU – 574146



**MICROSERVICES ARCHITECTURE AND
IMPLEMENTAION
MINIPROJECT REPORT
ON**

“STUDENT PLACEMENT PORTAL”

Submitted in the partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Miniproject entitled “Student Placement Portal” is a bonafide work carried out by *Ananya, Ananya v Naik*, bearing the USN *01SU23CS023,01SU23CS025*, in the partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering** of the **Srinivas University, Institute of Engineering & Technology** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Miniproject report has been approved as it satisfies the academic requirements in respect of practical work prescribed for the said degree.

Mr.Yogesh Bandewar
(Miniproject Guide)

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Department of Computer Science and Engineering

DECLARATION

We, **Ananya ,Ananya V Naik** the students of Sixth semester, **B.Tech** in Computer Science and Engineering, Srinivas University, Mukka, hereby declare that the Miniproject entitled “ Student Placement Portal” has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology in Computer Science and Engineering of Srinivas University, Institute of Engineering and Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: 27/03/2026

Place: Mukka

ABSTRACT

The **Student Placement Portal** is a web-based system designed to simplify and automate the placement process in educational institutions. It provides a centralized platform where students, recruiters, and administrators can interact efficiently.

The portal abstracts the complex placement workflow by dividing it into key modules such as student registration, company registration, job postings, application management, and result tracking. Students can create profiles, upload resumes, and apply for jobs based on their eligibility. Recruiters can post job opportunities, view student profiles, and shortlist candidates. Administrators manage the entire system, verify data, and coordinate placement activities.

The system hides the internal complexities like database management, communication handling, and eligibility filtering, presenting users with a simple and user-friendly interface. It ensures transparency, reduces manual work, saves time, and improves the overall efficiency of the placement process.

In short, the portal provides an abstract view of the placement system by focusing only on essential functionalities while concealing the underlying technical details

The **Student Placement Portal** is an integrated digital platform that automates and manages the end-to-end placement process in an educational institution. The system is designed using the principle of **abstraction**, where only the necessary functionalities are exposed to users while hiding the internal implementation details.

At a higher level, the portal consists of three main user roles: **Students, Recruiters, and Administrators**. Each role interacts with the system through a simplified interface without needing to understand the backend processes such as database operations, authentication mechanisms, or eligibility algorithms.

The system abstracts the placement workflow into several functional modules:

- **StudentModule:**

Allows students to register, create and update profiles, upload resumes, and apply for job opportunities. It filters jobs based on eligibility criteria such as CGPA, skills, and branch, hiding complex filtering logic.

- **RecruiterModule:**

Enables companies to register, post job openings, specify eligibility requirements, and shortlist candidates. The system abstracts candidate matching and application tracking processes.

- **AdminModule:**

Acts as the control unit of the portal. Administrators verify student and company data, manage job postings, monitor applications, and schedule recruitment drives. All system-level operations are simplified into manageable controls.

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- ANANYA

-ANANYA V NAIK

CHAPTER 1: INTRODUCTION

1.1 Overview Of The Project

The **Student Placement Portal** is a full-stack web-based system developed to simplify and digitize the process of campus placements. The application provides an interactive platform where students can register, create profiles, upload resumes, and apply for job opportunities. It aims to offer a convenient and efficient alternative to traditional placement management methods.

The system is built using modern web technologies, combining a frontend interface with a backend server and a database. The frontend allows students, administrators, and recruiters to interact with the application, while the backend handles data processing, job postings, application management, and communication with the database. The database stores information about students, companies, and job openings, ensuring dynamic and organized data a management.

The application also demonstrates key concepts of full-stack development such as RESTful API integration, client-server architecture, and data handling. Features like job notifications, application tracking, resume management, and recruiter shortlisting make the system practical and user-friendly.

Overall, this project provides a foundation for real-world placement management systems and can be further enhanced with advanced features such as user authentication, interview scheduling, and performance analytics.

1.2 Problem Definition

In today's competitive environment, traditional methods of managing campus placements, such as manual record-keeping and offline communication, are time-consuming and inefficient. Students often face difficulties in accessing job opportunities, tracking application status, and receiving timely updates from the placement cell. Additionally, there is a lack of a centralized system to manage student data, company information, and job postings, which leads to poor organization and increased chances of errors.

1.3 Motivation

The motivation behind developing the **Student Placement Portal** arises from the increasing need for a structured and efficient system to manage campus placements. With the growing number of students and companies participating in recruitment drives, there is a demand for a platform that

provides quick and easy access to job opportunities without relying on manual processes.

Traditional placement methods are often time-consuming, prone to errors, and lack proper coordination between students, administrators, and recruiters. These limitations highlight the need for an automated system that can handle student registration, job postings, and application tracking in a seamless manner. The project aims to address these challenges by providing a user-friendly interface and efficient backend functionality.

Additionally, this project serves as a practical implementation of full-stack development concepts. It helps in understanding how frontend, backend, and database systems work together to build real-world applications. The motivation also includes improving technical skills, gaining hands-on experience, and developing a scalable solution that can be enhanced with advanced features such as resume analysis, interview scheduling, and real-time notifications in the future.

1.4 Objectives

The primary objective of this project is to design and implement an efficient **Student Placement Portal** that allows students to explore job opportunities and apply for placements online with ease. The system should provide a smooth and interactive experience, enabling students to create profiles, upload resumes, and track their applications conveniently.

Another objective is to ensure accurate handling of student and recruitment data. The system should properly manage student records, job postings, and application details, while dynamically updating information and handling user actions without errors or delays.

The project also aims to provide a user-friendly interface that simplifies the placement process. Students, administrators, and recruiters should be able to easily navigate through the application, view relevant information, and perform their tasks without complexity.

Additionally, the system should enable efficient backend management of placement activities through APIs. Administrators should be able to add, update, or delete student and company data, post job opportunities, and ensure smooth communication between the frontend and backend.

Finally, the objective is to gain practical experience in full-stack web development by integrating frontend, backend, and database technologies. This helps in enhancing skills in programming, API development, and data management, which are essential for building real-world web applications.

CHAPTER 2: LITERATURE REVIEW

2.1 Review and Limitations of Existing Systems

Existing placement management systems in many institutions are often based on traditional methods such as manual record-keeping, notice boards, and offline communication. While these methods are widely used, they come with several challenges. Managing placement activities manually requires more time, may lead to miscommunication between students and placement officers, and depends heavily on human effort, making the overall process less efficient.

The main limitations of these systems include lack of convenience, limited accessibility, and higher chances of errors. Students often have to rely on physical notices or informal communication to get information about job opportunities, which can be time-consuming and unreliable. There is also a possibility of missing important updates or deadlines due to improper communication.

Additionally, these systems do not provide real-time updates, centralized data management, or proper tracking of student applications. Maintaining records manually becomes difficult, especially when dealing with a large number of students and companies. There is no efficient way to filter eligible candidates or analyze placement data, which reduces overall effectiveness.

Therefore, traditional placement systems are not suitable for today's fast-paced and digital environment, highlighting the need for an automated and web-based Student Placement Portal that ensures efficiency, accuracy, and better communication.

2.2 Proposed System Overview

The proposed system is a **Student Placement Portal**, a web-based application that allows students to explore and apply for job opportunities online through a centralized platform. The system includes features such as student registration, profile creation, resume upload, job listings, and application management through an interactive interface.

When a user accesses the application, data related to students, companies, and job opportunities is fetched from a centralized database and displayed on the screen. Students can view available jobs, apply for suitable positions, and track their application status easily. Administrators can manage student data, post job openings, and monitor placement activities efficiently.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Overview

The system architecture for the **Student Placement Portal** is based on a client-server model, consisting of three main components: the user interface (frontend), the application server (backend), and the database. The frontend provides an interface where students, administrators, and recruiters can interact with the system, such as registering, viewing job opportunities, uploading resumes, and applying for jobs. The database stores student details, company information, and job postings securely.

The workflow of the system follows a structured process: users access the application through a web browser, create profiles, and explore available job opportunities. The frontend sends requests to the backend, which processes the data and retrieves relevant information from the database.

User Input → Request to Backend → Data Processing → Database Interaction → Response Display

3.2 Hardware Requirements

- Processor: Intel i3 / i5 / AMD equivalent or higher
- RAM: Minimum 4 GB (8 GB recommended for better performance)
- Hard Disk: Minimum 20 GB free space
- Input Devices: Keyboard and Mouse
- Output Devices: Monitor
- Network: Stable internet connection for accessing the system

3.3 Software Requirements

- Operating System: Windows, Linux, or macOS
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technologies: Node.js / Express.js
- Database: MongoDB for storing food item data

CHAPTER 4: METHODOLOGY

4.1 Working Principle

The **Student Placement Portal** works based on a client-server architecture where users interact with the system through a web interface. Initially, students, administrators, and recruiters access the application and perform actions such as registration and login. The frontend communicates with the backend server to fetch and display relevant data such as student profiles, job opportunities, and company details from the database.

Once the data is loaded, students can view available job listings, check eligibility criteria, and apply for suitable positions by submitting their resumes. The frontend communicates with the backend server to fetch and display relevant data such as student profiles, job opportunities, and company details from the database. Administrators can manage student records, post job openings, and monitor placement activities, while recruiters can view applications and shortlist candidates based on their requirements.

The updated information, such as application status or job updates, is then sent back to the frontend and displayed to the user. The system ensures proper data handling, maintains accuracy, and provides real-time updates to improve the overall placement process. The data is loaded, students can view available job listings, check eligibility criteria, and apply for suitable positions by submitting their resumes. The frontend communicates with the backend server to fetch and display relevant data such as student profiles, job opportunities, and company details from the database. Administrators can manage student records, post job openings, and monitor placement activities, while recruiters can view applications and shortlist candidates based on their requirements.

The system ensures proper data handling, maintains accuracy, and provides real-time updates to improve the overall placement process. The updated information, such as application status or job updates, is then sent back to the frontend and displayed to the user. The system ensures proper data handling, maintains accuracy, and provides real-time updates to improve the overall placement process.

4.2 Module Description

1. **InputModule:**

Accepts user actions such as registration, login, viewing job listings, and applying for jobs. It captures user inputs and sends requests to the backend for processing.

2. **Authentication Module(Optional):**

Verifies user identity by checking login credentials (username and password) and ensures that only authorized users (students, admins, recruiters) can access specific features of the system.

3. **Student Management Module:**

Displays Manages student-related data such as profile details, academic information, and resumes. It allows students to create, update, and maintain their profiles in the system.

4. **Application Module:**

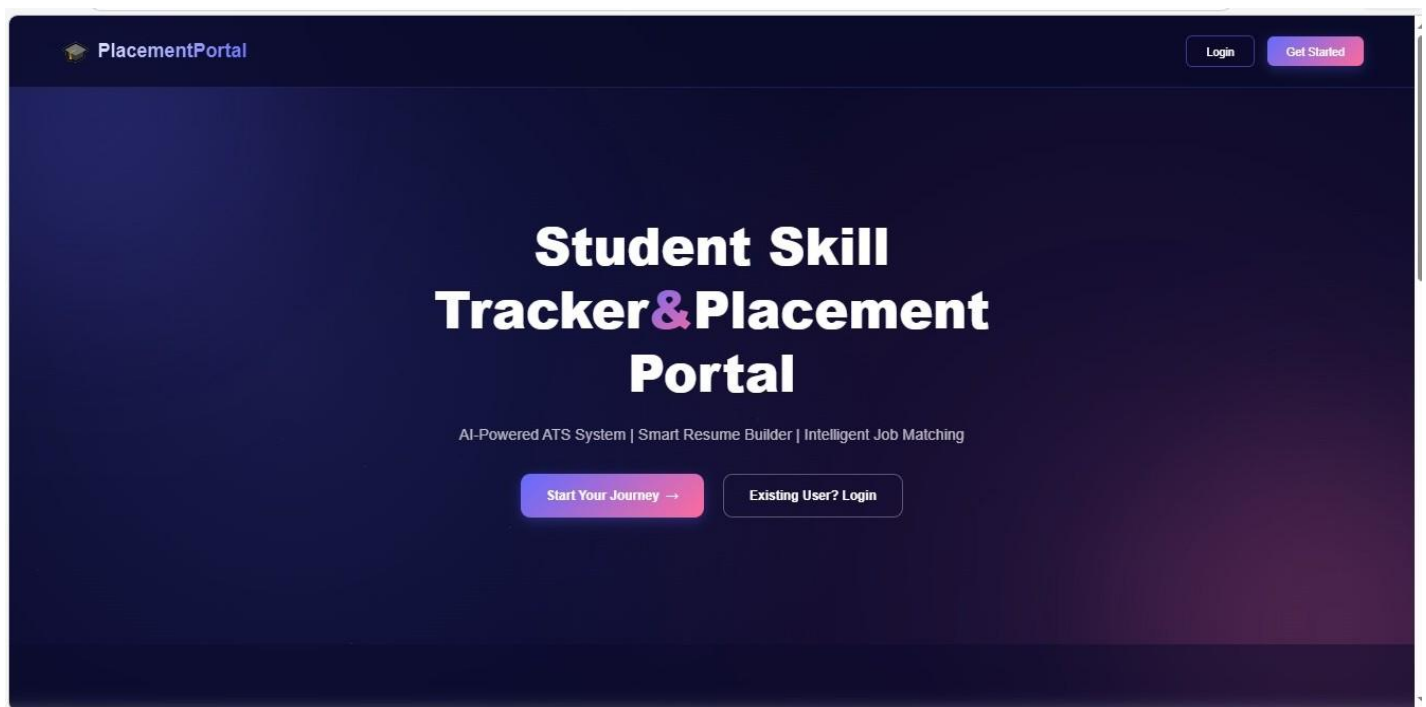
Allows Allows students to apply for jobs, submit resumes, and track their application status. It also enables recruiters to view applications and shortlist candidates.

5. **OutputModule:**

Displays results such as job listings, application status, and notifications. It ensures that users receive proper feedback and updates for their actions within the system.

CHAPTER 5: RESULTS

5.1 Output



Create Account

Join the Placement Portal

Full Name

Email

Password

Confirm Password

I am a

Student

Register

Already have an account? [Login here](#)

Student Dashboard

Build ResumeLogout

A

ananya

ananya@gmail.com

Student

50%

Profile Complete

30%

Placement Ready

0

Skills

Profile Completion

50%

Education

College	Branch	CGPA
SUIET	CSE	9

Skills & Proficiency

No skills added yet. Click Edit Profile to add skills.

Social Links

GitHub: Not added

LinkedIn: Not added

 Student Dashboard

Build ResumeLogout

 Available Jobs

full stack develooper

 amazon •  mangaluru

 50000

 Required: mongo db, react js, html

 Deadline: 27/3/2026

Apply Now →

Backend Developer

 Amazon •  Chennai

 16-20 LPA

 Required: React js

 Deadline: 31/3/2026

Apply Now →

Full Stack Developer

 IBM •  Bangalore

 15-20 LPA

 Required: React, Nodejs, MongoDB

 Deadline: 31/12/2025

Apply Now →

Welcome Back

Login to your account

Email

Password

Login

Don't have an account? [Register here](#)

Recruiter DashboardLogout

[+ Post New Job](#)

Your Job Postings

full stack developer
amazon • mangaluruView Applicants (0)

🏷️ 50000 🛠️ Skills: mongo db, react js, html

Backend Developer
Amazon • ChennaiView Applicants (0)

🏷️ 16-20 LPA 🛠️ Skills: React.js

Full Stack Developer
IBM • BangaloreView Applicants (0)

Backend Developer
Amazon • ChennaiView Applicants (0)

🏷️ 16-20 LPA 🛠️ Skills: React.js

Full Stack Developer
IBM • BangaloreHide Applicants (2)

🏷️ 15-20 LPA 🛠️ Skills: React, Node.js, MongoDB

Applicants (2)

Student	ATS Score	Status	Resume	Actions
69c26c21c924f0d38a8bf05b	10%	APPLIED	View Resume	Shortlist Select Reject
69bff95e61bc083c7aed722f	70%	SELECTED	View Resume	Shortlist Reject

Challenges

One of the major challenges was handling user interactions efficiently. Ensuring that user actions such as registration, login, and applying for jobs worked correctly required careful implementation of frontend and backend communication. Initial issues like incorrect application submissions or delays in updates were resolved by improving state management and API handling.

Another challenge was managing placement data dynamically. It was important to fetch, display, and update job postings and student information from the database without errors. Handling this required proper API design and testing to ensure smooth data flow between the server and the user interface.

Database management was also a challenge. Designing the structure to store student details, company information, and job data in an organized manner required careful planning. Issues like incorrect queries and data inconsistency were resolved by optimizing database operations.

Handling errors and unexpected user inputs was another difficulty. The system initially produced errors when invalid login credentials were entered or when backend routes were not properly defined. These issues were resolved by adding proper validation, routing, and error-handling mechanisms.

The system ensures accuracy in placement management by reducing human errors that commonly occur in manual processes. Students can clearly view job details such as eligibility criteria, job role, and company information before applying, minimizing confusion and mistakes.

Ensuring smooth communication between frontend and backend was also challenging. Problems related to API integration and asynchronous requests were addressed through debugging and proper handling of responses.

Finally, testing the application under different scenarios such as multiple user actions and edge cases required extra effort. By thoroughly testing the system, the reliability and performance of the Student Placement Portal were improved.

improved.

CHAPTER 6: CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion

The **Student Placement Portal** was successfully designed and implemented as a web-based system to provide an efficient and user-friendly platform for managing campus placements. The project demonstrated how modern web technologies and database management can be used to simplify the placement process. It allows students to view job opportunities, apply for positions, and track their application status while ensuring accurate data handling.

Moreover, the project provided practical exposure to real-world application development, including frontend design, backend processing, and database integration. It helped in understanding important concepts such as API development, data management, and client-server architecture. By testing the system under different scenarios, its reliability and performance were ensured. Overall, the project enhanced technical skills, problem-solving ability, and knowledge of developing efficient and user-friendly web applications.

6.2 Future Enhancements

In the **Student Placement Portal** can be enhanced by implementing advanced user authentication features such as secure login, role-based access, and profile verification to provide personalized experiences. It can also be upgraded to support large-scale usage with improved performance and scalability.

Additional features such as interview scheduling, real-time notifications, and email alerts can be added to improve communication between students, administrators, and recruiters. A mobile application version of the system can also be developed to make access more convenient and user-friendly.

Furthermore, advanced features like resume analysis, skill-based job recommendations, and automated candidate shortlisting can be integrated to improve efficiency. The system can also include admin dashboards, detailed analytics, and reporting tools for better placement management. Finally, integrating cloud-based infrastructure can improve system availability, reliability, and performance for handling a large number of users effectively.

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